Sakuna Harinda Jayasundara

Software Engineer & Researcher

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Professional Summary

Software engineer and researcher with 5+ years of combined industry and research experience building scalable, production-grade systems across backend development, machine learning, and automation. Brings 8+ years of experience in Python, Java, and C/C++ development, along with extensive work across agentic AI, agentic security, modern ML and MLOps frameworks, continuous integration and delivery (CI/CD) pipelines, and test automation. Ph.D. from the University of Auckland, with research spanning deep learning, natural language processing, cybersecurity, and human-computer interaction (HCI). Focused on applying data-driven and research-informed approaches to solve real-world engineering problems.

Technical Skills

AI Engineering & Agentic Systems: LangGraph, CrewAI, AutoGen, Google ADK, MCP, RAG, AI Evaluation, AI Agents, Agentic AI security, Microsoft Agent Governance Toolkit

Languages: Python, Java, C/C++, Go, C#/.NET, TypeScript, Dart, CSS

Backend & APIs: Spring Boot, FastAPI, REST, SOAP, JSON-RPC, microservices, Cloudflare

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn; model fine-tuning, from-scratch architecture implementation (reimplementing layers from papers), natural language processing (NLP), computer vision

Data: SQL (PostgreSQL), MongoDB, Neo4j, Neon, Supabase/Firebase

Cloud & DevOps/MLOps: AWS, Azure, Docker, Kubernetes, Jenkins, MLflow, Airflow, GitHub Actions, Git, Graph API

AI Dev Tools: Claude, GitHub Copilot/Copilot Studio, Cursor

Experience

Post-doctoral Research Fellow *Oct 2026 – Present*

University of Auckland | Auckland, NZ

- Investigate agentic AI governance and security, developing mechanisms for runtime policy enforcement, least-privilege access, and secure delegation to constrain autonomous agent actions.
- Research online provenance-aware threat detection and adaptive access-control policy generation for multi-agent AI systems, integrating provenance graphs, behavioral analysis, and validated policy synthesis to detect threats and mitigate unauthorized information flows.
- Develop security mechanisms for agentic systems, addressing prompt injection, unauthorized tool use, privilege escalation, and inter-agent trust through provenance-grounded threat analysis and policy-enforced execution.

Doctoral Researcher (AI and cybersecurity) *Oct 2022 – Jan 2026*

University of Auckland | Auckland, NZ

- Designed transformer-based ML pipelines that convert natural-language requirements into structured, validated access control policies, reaching approximately 80% F1-score through context retrieval, generation, validation, and human-in-the-loop feedback.
- Fine-tuned and continually pre-trained off-the-shelf language models and built deep learning architectures from scratch in PyTorch, including reimplementing model layers directly from scientific papers where no public implementation existed.
- Designed ML-based mechanisms that support automatic error detection and recovery, enabling safe deployment of AI-generated security policies in high-risk environments.
- Conducted empirical user studies with 19 professional policy implementers, demonstrating improved accuracy, efficiency, trust, and adoption potential compared to existing approaches.
- Produced comprehensive analyses and datasets that address long-standing research gaps in access control policy generation.

Professional Teaching Fellow *Nov 2022 – Present*

University of Auckland | Auckland, NZ

- Prepare and deliver lectures and tutorials for undergraduate and postgraduate courses, including Data Structures & Algorithms, Software Engineering, Cryptographic Management, and Java Programming for Industry.
- Designed course materials and built autograded programming assignments with interactive learning tools, supporting consistent assessment across large student cohorts (approximately 300).

- Supervise examinations and assess assignments and projects, ensuring fair and consistent evaluation.

Software Engineer (SDET) *Jun 2022 – Oct 2022*

H2O.ai | Remote

- Designed and implemented end-to-end test automation pipelines using Python, Jenkins, and Groovy, reducing manual testing effort and increasing release velocity roughly by 30%.
- Led the architecture of a reusable test automation framework, defined the coding standards, and shaped the automation roadmap for cross-team use.
- Designed, developed, and maintained web applications supporting internal and external ML workflows.
- Collaborated with engineering teams to integrate automation into CI/CD workflows, improving code quality and shortening time-to-deployment for key features.

Software Engineer *Jul 2021 – Jun 2022*

Axiata Digital Labs | Colombo, Sri Lanka

- Designed, developed, and maintained backend RESTful and SOAP APIs in Java and Spring Boot for customer-facing services tied to Celcom Malaysia, ensuring high reliability and clean integration with existing telecommunications systems.
- Designed an anomaly detection system with OpenTelemetry and ML models, enhancing production observability and enabling the early detection of 300+ critical system anomalies.
- Developed an NLP-based repository classifier to automatically surface relevant public GitHub projects, reducing manual competitive-analysis workload.

Intern Electronics Engineer *May 2019 – Dec 2019*

Paraqum Technologies | Colombo, Sri Lanka

- Developed a GTP (GPRS Tunnelling Protocol) packet analysis tool and test environment for inspecting and debugging mobile core network traffic.
- Built a load-balancing module to monitor network interfaces and manage traffic; optimized existing C++ modules for responsiveness and reliability.

Projects

Marigold — Wedding Planning Platform *2026 – Present*

Full-stack wedding planning web application designed and built end to end with AI-led development, taking it from concept and design through to a working, deployed product. *(Next.js, Neon, Clerk, Cloudflare, Playwright, Astro, React, Tailwind CSS)*

Helixar.ai — Agentic AI Security Platform *2026 – Present*

Contributing to Helixar, a control plane for enterprise AI agents delivering governance, compliance, and runtime security, including agent identity and scope enforcement, behavioural monitoring, and tamper-evident audit and delegation provenance for autonomous LLM agents. *(Python, TypeScript, LangChain, MCP, LLM agents)*

Ragatouille — Open-Source RAG Guide *2024*

Comprehensive open-source guide to building Retrieval-Augmented Generation systems with Langgraph and Neo4j. **Featured on LangChain's official LinkedIn page (Link).** *(Python, Langgraph, Neo4j, NLP)*

Education

Ph.D. in Computer Science *Oct 2022 – Jan 2026*

University of Auckland | Auckland, NZ

- Researched human-centric automated access control policy generation using large language models.
- Four publications (two flagship cybersecurity conferences, two Q1 journals), two under review. Full list on Google Scholar.

B.Sc. (Hons.) in Electronics & Telecommunication Engineering *2016 – 2021*

University of Moratuwa | Colombo, Sri Lanka

- Attained a GPA of 3.92/4.2, obtaining a first class.
- Dean's List Honouree, recognised in all four academic years of the B.Sc. program.
- Developed a deep learning-based maritime surveillance software for the Sri Lanka Navy to detect suspicious activities in the sea using PyTorch and PyQt. The project won the 2nd runner-up in the National ICT Awards for the best university project.

Honours & Awards

- ESET Chillisoft Scholarship in Cybersecurity (2022)
- MBIE Artificial Intelligence for Human-Centric Security Project Scholarship (2022)
- 2nd Runner-Up, National ICT Awards — Best University Project (2021)